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To: The Editorial Board, *Nature Machine Intelligence / Cell Systems*

Subject: Submission of Original Research Article: *The Instruction Set Architecture of Living Genomes*

Dear Editors,

I am pleased to submit our original research manuscript, '**The Instruction Set Architecture of Living Genomes: Universal Biological Logic Gates, Wobble Carrier Waves, and De Novo Dual-Phase Compilation**', for consideration as a Research Article.

For more than seven decades, molecular biology has operated under the conceptual framework that nucleic acids function as linear, single-track text. In this work, we present a unified computational and information-theoretic framework demonstrating that biological genomes operate as multi-threaded machine code executing on a physical molecular virtual machine:

- **The Wobble Carrier Wave Theorem:** Shannon entropy across codon positions proves that third-position wobble nucleotides ($H = 1.984$ bits) serve as an orthogonal, high-bandwidth carrier wave transmitting secondary (+1/-1 frame) protein instructions with zero non-synonymous penalty to the primary frame.
- **Universal Hardware Logic Density (~10.14 gates/kb):** We present the largest multi-epoch census of biological logic switches to date, identifying **99,348 hardware logic gates across 10.17 million base pairs**, proving this density was already operational 3.8 billion years ago in Archean extremophiles.
- **De Novo Dual-Phase Recompilation:** We demonstrate synthesizing a single physical DNA strand that compresses two disparate therapeutic and antiviral proteins with 1.98x physical information compression.

This manuscript provides the mathematical foundations and open-source software tools for reverse-engineering biological machine code and designing high-density gene therapies. All code and pipelines are 100% reproducible and archived under CERN Zenodo DOI 10.5281/zenodo.22147682.

Thank you for your time and peer review.

Sincerely,

Jason Rezek